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LeanForControl

Anand Gokhale

Titus Quah

Claude Code

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Chapter 1

Linear systems

1.1 Observability and controllability matrices

For a finite-dimensional linear system

$$\dot{x} = Ax + Bu, \quad y = Cx,$$

with state matrix A , input matrix B , and output matrix C , this section collects the (finite-horizon) controllability and observability matrices and the bridge theorem connecting the textbook observability predicate to the kernel of the observability matrix.

1.1.1 Observability matrix

Definition 1. `LinearSystems.observabilityMatrix` The *observability matrix* of a pair (A, C) with $A \in \mathbb{F}^{n \times n}$ and $C \in \mathbb{F}^{p \times n}$ is the block-row matrix

$$\mathcal{O}(A, C) = \begin{bmatrix} C \\ C A \\ C A^2 \\ \vdots \\ C A^{n-1} \end{bmatrix} \in \mathbb{F}^{(np) \times n}.$$

Rows are indexed by `Fin n × Fin p`, so that A^k is available without casting $k : \text{Fin } n$ through `Fin.val`.

Definition 2. `LinearSystems.IsObservable` A linear system (A, C) is *observable* when the only state $x \in \mathbb{F}^n$ for which

$$C A^k x = 0 \quad \text{for every } k = 0, 1, \dots, n-1$$

is the zero state. This phrasing does not mention $\mathcal{O}(A, C)$, so the bridge ?? has real content.

Theorem 3. *def:observabilityMatrix LinearSystems.observabilityMatrix_applyBlock-rowentryshape : atrow (k, i) and column j , the observability matrix coincides with the (i, j) entry of $C \cdot A^k$:*

$$\mathcal{O}(A, C)_{(k,i),j} = (C \cdot A^k)_{i,j}.$$

Proof. Holds definitionally by the encoding chosen for ??.

□

Theorem 4. *def:observabilityMatrix LinearSystems.observabilityMatrix_mulVec_applyForEveryState $x \in \mathbb{F}^n$ and every $(k, i) \in \text{Fin } n \times \text{Fin } p$,*

$$(\mathcal{O}(A, C) \cdot x)_{(k,i)} = (C \cdot A^k \cdot x)_i.$$

This is the definitional bridge that powers ??.

Proof. Holds by definitional unfolding of $\mathcal{O}(A, C)$ (see ??).

□

1.1.2 Controllability matrix

Definition 5. *LinearSystems.controllabilityMatrix* The *controllability matrix* of a pair (A, B) with $A \in \mathbb{F}^{n \times n}$ and $B \in \mathbb{F}^{n \times m}$ is the block-column matrix

$$\mathcal{C}(A, B) = [B \quad AB \quad A^2B \quad \dots \quad A^{n-1}B] \in \mathbb{F}^{n \times (nm)}.$$

Columns are indexed by $\text{Fin } n \times \text{Fin } m$, so that A^k is available without casting $k : \text{Fin } n$ through Fin.val .

Theorem 6. *def:controllabilityMatrix LinearSystems.controllabilityMatrix_applyBlock-columnentryshape : atrow i and column (k, j) , the controllability matrix coincides with the (i, j) entry of $A^k B$:*

$$\mathcal{C}(A, B)_{i,(k,j)} = (A^k B)_{i,j}.$$

Proof. Holds definitionally by the encoding chosen for ??.

□

Definition 7. *LinearSystems.IsControllable* A linear system (A, B) is *controllable* when every target state $x \in \mathbb{F}^n$ is reachable from the origin in n steps: there exist input vectors $u_0, u_1, \dots, u_{n-1} \in \mathbb{F}^m$ such that

$$x = \sum_{k=0}^{n-1} A^k B u_k.$$

This phrasing does not name the controllability matrix, so the bridge ?? has real content.

1.1.3 Milestone: observability iff trivial kernel of $\mathcal{O}$

Theorem 8. *def:observabilityMatrix, def:isObservable LinearSystems.isObservable_iff_observabilityMatrix_kernel_dimensional_linear_system (A, C) is observable in the sense of ?? if and only if the observability matrix $\mathcal{O}(A, C)$ has trivial kernel under matrix-vector multiplication : $\text{IsObservable}(A, C) \iff (\forall x, \mathcal{O}(A, C) \cdot x = 0 \Rightarrow x = 0)$.*

Proof. `lem:observabilityMatrix-mulVec-apply` Both directions follow from the bridge `??`, which identifies the (k, i) -coordinate of $\mathcal{O}(A, C) \cdot x$ with the i -coordinate of $(C A^k) \cdot x$. The forward direction reads the per-power kernel out of the assembled kernel coordinatewise; the reverse direction packs them back. $\square$

1.1.4 Rank characterizations

Theorem 9. `def:observabilityMatrix, def:isObservable LinearSystems.isObservable` *if f_o bservabilityMatrix, rank dimensionalsystem (A, C) is observable if and only if the observability matrix $\mathcal{O}(A, C)$ has full column rank, i.e. $\text{rank } \mathcal{O}(A, C) = n$, where n is the state dimension.*

Proof. `thm:isObservable-iff-ker-trivial` Chain the kernel-form milestone `??` with the matrix-level bridge $(\forall x, M \cdot x = 0 \Rightarrow x = 0) \iff \text{rank } M = n$ from ‘MatrixLemmas’, applied with $M = \mathcal{O}(A, C)$. $\square$

Theorem 10. `def:controllabilityMatrix, def:isControllable LinearSystems.isControllable` *if f_c ontrollabilityMatrix (A, B) is controllable if and only if the controllability matrix has full row rank, $\text{rank } \mathcal{C}(A, B) = n$, where n is the state dimension.*

Proof. Combine the reindexing identity $\mathcal{C}(A, B) \cdot u = \sum_k A^k B \cdot u_k$ (with $u_k(j) = u(k, j)$) with the matrix-level bridge $(\forall y, \exists x, M x = y) \iff \text{rank } M = n$ from ‘MatrixLemmas’, applied with $M = \mathcal{C}(A, B)$. $\square$

1.1.5 Hautus observability

Definition 11. `LinearSystems.unobservableSubspace` The *unobservable subspace* of (A, C) is the A -invariant subspace

$$\mathcal{N}(A, C) = \{v \in \mathbb{C}^n : C A^k v = 0 \text{ for every } k = 0, 1, \dots, n-1\}.$$

Equivalently, $\mathcal{N}(A, C) = \ker \mathcal{O}(A, C)$, but here it is phrased without naming $\mathcal{O}$ so the bridge `??` reads as content.

Theorem 12. `def:unobservableSubspace, def:isObservable LinearSystems.unobservableSubspace` *if f_i sObservable dimensionalsystem (A, C) is observable in the sense of `??` if and only if its unobservable subspace is trivial : $\mathcal{N}(A, C) = \{0\} \iff \text{IsObservable}(A, C)$.*

Proof. Both directions are membership unfoldings of `??` against `??`. $\square$

Theorem 13. `def:unobservableSubspace LinearSystems.A_mulVec_m em_u nobservableSubspace_o f_m em` *The unobservable subspace $\mathcal{N}(A, C)$ is A -invariant, i.e. for every $v \in \mathcal{N}(A, C)$, also $A v \in \mathcal{N}(A, C)$.*

Proof. For $k = 0, \dots, n-2$ this is direct from the definition. For $k = n-1$ we land at $C A^n v$, which Cayley-Hamilton rewrites as a $\mathbb{C}$ -linear combination of $C A^i v$ for $i < n$. Each of those is zero, so the combination is zero. $\square$

Definition 14. `LinearSystems.hautusObservabilityMatrix` The *Hautus observability matrix* of (A, C) at a complex number μ is the block-row matrix

$$H_{A,C}(\mu) = \begin{bmatrix} \mu I - A & C \end{bmatrix} \in \mathbb{C}^{(n+p) \times n}.$$

Theorem 15. *def:hautusObservabilityMatrix LinearSystems.hautusObservabilityMatrix_mulVec_eq_zero_iffFore_e C and $v \in \mathbb{C}^n$,*

$$H_{A,C}(\mu) \cdot v = 0 \iff (Av = \mu v \wedge Cv = 0).$$

Proof. □

Theorem 16. *def:isObservable,def:hautusObservabilityMatrix LinearSystems.not_isObservable_iimplies_hautus_f C is not observable, then there exists $\mu \in \mathbb{C}$ and a nonzero vector $v \in \mathbb{C}^n$ such that*

$$H_{A,C}(\mu) \cdot v = 0,$$

i.e. μ is an eigenvalue of A with an eigenvector in $\ker C$.

Proof. *def:unobservableSubspace,lem:hautus-mulVec-eq-zero-iff,lem:unobservableSubspace-invariant,thm:unobservable-eq-bot-iff-observable* The unobservable subspace ?? is nontrivial by ?. It is A -invariant by ?, hence finite-dimensional and invariant; over $\mathbb{C}$ this guarantees a nonzero eigenvector inside it. That vector is automatically in $\ker C$ (membership at $k = 0$), so it lies in the kernel of $H_{A,C}(\mu)$ via ?. □

Theorem 17. *def:isObservable,def:hautusObservabilityMatrix LinearSystems.hautus_failure_iimplies_not_isObservable_i C and $v \neq 0$ satisfy $H_{A,C}(\mu) \cdot v = 0$, then (A, C) is not observable.*

Proof. *lem:hautus-mulVec-eq-zero-iff* The witness gives $Av = \mu v$ and $Cv = 0$. By induction on k , $A^k v = \mu^k v$, so $CA^k v = \mu^k Cv = 0$ for every k . The vector v is therefore in the unobservable subspace and is nonzero, so observability fails. □

Theorem 18. *def:isObservable,def:hautusObservabilityMatrix LinearSystems.isObservable_iff_hautus_Afinite-dimensional system (A, C) over $\mathbb{C}$ is observable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H_{A,C}(\mu)$ has trivial kernel:*

$$\text{IsObservable}(A, C) \iff \forall \mu \in \mathbb{C}, \ker H_{A,C}(\mu) = \{0\}.$$

Proof. *thm:not-isObservable-implies-hautus-failure,thm:hautus-failure-implies-not-isObservable* Combine ? (failure direction) and ? (converse). □

1.1.6 Hautus controllability

Definition 19. *LinearSystems.hautusControllabilityMatrix* The *Hautus controllability matrix* of (A, B) at a complex number μ is the block-column matrix

$$H_{A,B}^{\text{ctrl}}(\mu) = [\mu I - A \quad B] \in \mathbb{C}^{n \times (n+m)}.$$

Theorem 20. *def:isControllable,def:hautusControllabilityMatrix LinearSystems.isControllable_iff_hautus_Afinite-dimensional system (A, B) over $\mathbb{C}$ is controllable if and only if for every $\mu \in \mathbb{C}$ the Hautus matrix $H_{A,B}^{\text{ctrl}}(\mu)$ has full row rank:*

$$\text{IsControllable}(A, B) \iff \forall \mu \in \mathbb{C}, \text{rank } H_{A,B}^{\text{ctrl}}(\mu) = n.$$

Proof. thm:isObservable-iff-rank,def:controllabilityMatrix,thm:isControllable-iff-rank,lem:observabilityMatrix-apply,def:observabilityMatrix,thm:isObservable-iff-hautus,def:isObservable,def:hautusObservabilityMatrix,lem:controllabilityMatrix-apply Combine the duality bridge ?? *nope, that's the iff itself*; actually combine *the duality bridge* **isControllable_iff_isObservable_transpose** with the observability Hautus iff ??, then convert the kernel-form RHS to the rank form via the matrix bridge **mulVec_kernel_trivial_iff_rank_eq_card_cols** from 'MatrixLemmas', and finally identify the transposed Hautus matrices. $\square$

Chapter 2

Lyapunov stability

- 2.1 Trajectories, equilibria, and comparison functions
- 2.2 Stability predicates
- 2.3 Sublevel sets and positive invariance
- 2.4 Lyapunov functions
- 2.5 Milestone theorems for autonomous systems